(MATH101, 102, 311, 312) Calculus and Analysis

Problems

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Contents

1 (MATH101) Calculus I

Problem 1.1. (2021 Spring KAIST Calculus I) Does each of the following infinite series converge? Mark Yes or No. You do not need to justify your answers.

(1) $\displaystyle\sum_{n=1}^{\infty} \frac{n!}{n^n}$

(2) $\displaystyle\sum_{n=1}^{\infty} \frac{e^{\sqrt{n}}}{n\sqrt{n}}$

(3) $\displaystyle\sum_{n=1}^{\infty} \left(1-\frac{1}{n}\right)^{n}$

(4) $\displaystyle\sum_{n=1}^{\infty} (-1)^n \left(1-\frac{1}{n}\right)^{n\ln n}$

(5) $\displaystyle\sum_{n=1}^{\infty} \left(1-\frac{1}{n}\right)^{2n\ln n}$

(6) $\displaystyle\sum_{n=1}^{\infty} \frac{1}{(n+1)\ln(n+1)}$

(7) $\displaystyle\sum_{n=1}^{\infty} \frac{1}{(n+1)\ln(n+1)\ln\ln(n+1)}$

(8) $\displaystyle\sum_{n=1}^{\infty} \frac{1}{(n+1)(\ln(n+1))^2}$

(9) $\displaystyle\sum_{n=1}^{\infty} \frac{1}{\tan\left(\arccos\left(\frac{1}{n+1}\right)\right)}$

(10) $\displaystyle\sum_{n=1}^{\infty} \frac{\sin(1/n)}{n^\alpha}, \qquad \alpha > 0$

Problem 1.2. (2021 Spring KAIST Calculus I) Find the area of the portion of the ellipse

$$9x^2 + 16y^2 = 144$$

between the y-axis and $x = 2$.

Problem 1.3. (2021 Spring KAIST Calculus I) Find the antiderivative of

$$\frac{1}{(x^2+x+1)(x^2+x+2)(x^2+x+3)}.$$

Problem 1.4. (2021 Spring KAIST Calculus I) Does the following improper integral converge?

$$\int_0^\infty \frac{\sin^3 x}{x}\, dx.$$

Problem 1.5. (2021 Spring KAIST Calculus I) Prove or disprove:

(1) If $\displaystyle\sum_{n=1}^{\infty} a_n$ converges, then $\displaystyle\sum_{n=1}^{\infty} a_n^2$ also converges.

(2) If both $\displaystyle\sum_{n=1}^{\infty} a_n$ and $\displaystyle\sum_{n=1}^{\infty} |b_n|$ converge, then $\displaystyle\sum_{n=1}^{\infty} a_n b_n$ also converges.

(3) Assume $a_n \neq 0$ for all n. If $\displaystyle\lim_{n\to\infty} a_n = 0$ and $\displaystyle\sum_{n=1}^{\infty} |\sin(a_n)|$ converges, then $\displaystyle\sum_{n=1}^{\infty} |a_n|$ converges.

Problem 1.6. (2021 Spring KAIST Calculus I) Evaluate

$$\int_1^\infty \left(2\sqrt{x^2-1} - 2x + \frac{1}{x}\right) \mathrm{d}x.$$

Problem 1.7. (2021 Spring KAIST Calculus I) Let f be a continuously differentiable (i.e., $f(x)$ is differentiable and $f'(x)$ is continuous) one-to-one function defined on $[a,b]$. Assume that $f'(x) \neq 0$ on $[a,b]$. Compute the following sum of two integrals:

$$\int_{f(a)}^{f(b)} f^{-1}(y)\ \mathrm{d}y + \int_a^b f(x)\ \mathrm{d}x.$$

Problem 1.8. (2021 Spring KAIST Calculus I) Let n be a positive integer and denote the following improper integral by a_n:

$$a_n = \int_0^1 (\ln x)^n\ \mathrm{d}x.$$

(a) Show that the improper integral a_n is convergent. Compute the value of a_n.

(b) Is the series

$$\sum_{n=1}^\infty \frac{1}{a_n}$$

absolutely convergent, conditionally convergent, or divergent? Explain the reason using appropriate tests.

(c) Is the series

$$\sum_{n=2}^\infty (-1)^n \frac{1}{\ln |a_n|}$$

absolutely convergent, conditionally convergent, or divergent? Explain the reason using appropriate tests.

Problem 1.9. (2021 Spring KAIST Calculus I) Show that the power series

$$\sum_{n=2}^\infty n(n-1)x^n, \qquad |x| < 1,$$

can be expressed as

$$\frac{x^2}{(1-x)^3}.$$

Problem 1.10. (2021 Spring KAIST Calculus I) Consider the standard ellipse C,

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1, \qquad a > b > 0.$$

Denote by e the eccentricity of C.

(a) Find an upper bound of the function

$$f(t) = \sqrt{1 - e^2 \sin^2 t}$$

using the first two terms of its binomial series expansion.

(b) Show that the perimeter (total arc length) of C is less than

$$2\pi a \left(1 - \frac{e^2}{4}\right).$$

Problem 1.11. (2021 Spring KAIST Calculus I) Consider the power series

$$\sum_{n=1}^{\infty} (-1)^n \frac{(x-1)^{2n+1}}{n2^n}.$$

(a) Find its interval of convergence.

(b) At each point in the interval of convergence, discuss its mode of convergence as absolute or conditional convergence together with proper reasoning.

Problem 1.12. (2021 Fall KAIST Calculus I) Is each of the following statements true or false? Mark T or F. You do not need to justify your answers.

(1) $f = O(g)$ implies $f = o(g)$ for functions f and g that are positive for x sufficiently large.

(2) $x(\ln x)^{2022} = o(x^{1.001})$.

(3) If $f'(a) \neq 0$, then there exists a $\delta > 0$ such that $f(x) \neq f(a)$ for all $x \neq a$ with $a - \delta < x < a + \delta$.

(4) $\displaystyle\int_2^\infty \frac{1 - 2^{-x}}{1 + \sqrt{x}}\, dx$ converges.

(5) $\displaystyle\int_1^6 \frac{1}{x - 4}\, dx$ converges.

(6) $\displaystyle\int_2^\infty \frac{1}{\ln x}\, dx$ converges.

(7) $\displaystyle\int_1^2 \frac{1}{x \ln x}\, dx$ converges.

(8) $\displaystyle\int_{-\infty}^\infty \frac{1}{e^x + e^{-x}}\, dx$ converges.

(9) $\displaystyle\int_{-\infty}^\infty \frac{2x}{x^2 + 1}\, dx$ converges.

(10) $\displaystyle\lim_{x\to\infty} \frac{\ln(x + 10^{2022})}{\ln x} = 1$.

Problem 1.13. (2021 Fall KAIST Calculus I) Evaluate

$$\int_1^\infty \left(2\sqrt{x^2 - 1} - 2x + \frac{1}{x}\right) dx.$$

Problem 1.14. (2021 Fall KAIST Calculus I) Let f be a continuously differentiable (i.e., $f(x)$ is differentiable and $f'(x)$ is continuous) one-to-one function defined on $[a, b]$. Assume that $f'(x) \neq 0$ on $[a, b]$. Compute the following sum of two integrals:

$$\int_{f(a)}^{f(b)} f^{-1}(y)\ \mathrm{d}y + \int_a^b f(x)\ \mathrm{d}x.$$

Problem 1.15. (2021 Fall KAIST Calculus I)

(a) Derive $\dfrac{\mathrm{d}}{\mathrm{d}x}(\operatorname{arctanh} x)$ for $|x| < 1$.

(b) Find $\displaystyle\int x \operatorname{arctanh} x\ \mathrm{d}x$.

Problem 1.16. (2021 Fall KAIST Calculus I) Let

$$\Gamma(n) = \int_0^\infty x^{n-1} e^{-x}\ \mathrm{d}x$$

for a positive integer n.

(a) Use a comparison test to show that $\Gamma(n)$ converges.

(b) Show that $\Gamma(n) = (n-1)!$.

Problem 1.17. (2021 Fall KAIST Calculus I) Let $w(t)$ be a differentiable function and let κ and ρ be positive constants.

(a) Solve

$$\frac{\mathrm{d}}{\mathrm{d}t} w(t) + \kappa w(t) - \rho = 0 \qquad \text{or} \qquad w' + \kappa w - \rho = 0$$

with the initial condition $w(0) = v_0$.

(b) By setting $w(t) = \dfrac{\mathrm{d}}{\mathrm{d}t} y(t)$, solve

$$\frac{\mathrm{d}^2}{\mathrm{d}t^2} y(t) + \kappa \frac{\mathrm{d}}{\mathrm{d}t} y(t) - \rho = 0 \qquad \text{or} \qquad y'' + \kappa y' - \rho = 0$$

with $y(0) = 0$ and

$$\left. \frac{\mathrm{d}}{\mathrm{d}t} y(t) \right|_{t=0} = v_0.$$

Problem 1.18. (2021 Fall KAIST Calculus I)

(1) Solve the equation

$$xy' + 2y = \frac{8x}{(x-1)(x+1)(x+2)}$$

for $x > 1$.

(2) Solve the initial value problem

$$y' = 2x\sqrt{1 + y^2}, \qquad y(0) = 0.$$

Problem 1.19. (2021 Fall KAIST Calculus I) Determine the radius and interval of convergence of the following series.

(1) $\displaystyle\sum_{n=1}^{\infty} \frac{1}{n} x^n$

(2) $\displaystyle\sum_{n=1}^{\infty} \frac{1}{n!} x^n$

(3) $\displaystyle\sum_{n=0}^{\infty} (-1)^n x^{2^n}$

Problem 1.20. (2021 Fall KAIST Calculus I) Consider the power series

$$\sum_{n=1}^{\infty} (-1)^n \frac{(x-1)^{2n+1}}{n2^n}.$$

(a) Find its interval of convergence.

(b) At each point in the interval of convergence, discuss its mode of convergence as absolute or conditional convergence together with proper reasoning.

2 (MATH102) Calculus II

Problem 2.1. (2021 Spring KAIST Calculus I) Mark True or False for each of the following statements. You do not need to justify your answers.

(1) If a series $\sum_{n=0}^{\infty} a_n x^n$ converges on $[-1, 1]$, then the series $\sum_{n=1}^{\infty} n a_n x^{n-1}$ converges on $(-1, 1)$.

(2) If $\mathbf{a}, \mathbf{b}, \mathbf{c}$ are three-dimensional vectors, then

$$\mathbf{a} \times (\mathbf{b} \times \mathbf{c}) = (\mathbf{a} \times \mathbf{b}) \times \mathbf{c}.$$

(3) If $\mathbf{a}, \mathbf{b}, \mathbf{c}$ are nonzero three-dimensional vectors and $(\mathbf{a} \times \mathbf{b}) \cdot \mathbf{c} = 0$, then $\mathbf{a}, \mathbf{b}, \mathbf{c}$ are in the same plane.

(4) Assume f is infinitely differentiable at 0 (i.e., $f^{(n)}(0)$ exists for all $n = 1, 2, 3, \ldots$). Then there exists $\varepsilon > 0$ such that

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} x^n$$

for all $-\varepsilon < x < \varepsilon$.

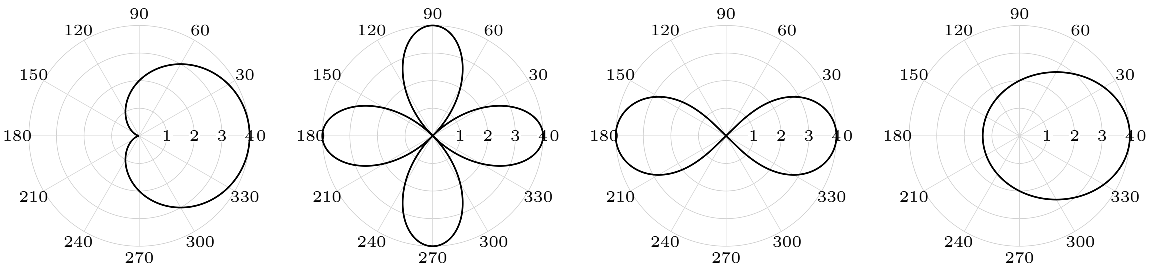
(5) Let $A(x) = \sum_n a_n x^n$ and $B(x) = \sum_n b_n x^n$ be power series with radii of convergence R_1 and R_2, respectively. Then $A(x)B(x)$ is again a power series with radius of convergence $\min\{R_1, R_2\}$.

Problem 2.2. (2021 Spring KAIST Calculus I) Match each of the following polar equations with an appropriate graph.

(1) $r = 4\cos(2\theta)$

(2) $r = \dfrac{4}{2 - \cos\theta}$

(3) $r = 2(1 + \cos\theta)$



(1) (2) (3) (4)

Problem 2.3. (2021 Spring KAIST Calculus I) Let

$$L_1: \ x = 1 + t, \ y = 3 - t, \ z = 2t, \qquad L_2: \ x = 0, \ y = -2s, \ z = 5s.$$

Find the minimum of $\|\overrightarrow{P_1P_2}\|$ for $P_1 \in L_1$ and $P_2 \in L_2$.

Problem 2.4. (2021 Spring KAIST Calculus I) Let M_1 be the plane in space given by

$$2x - y - 3z = 2,$$

and let M_2 be the plane perpendicular to M_1 through the points $P(1, -2, -4)$ and $Q(0, -1, -2)$.

(a) Find parametric equations for the line L of intersection of M_1 and M_2.

(b) Find an equation for a plane M parallel to L that contains the line through the points $A(-2, -4, 3)$ and $B(2, 0, 4)$. Find the distance between L and M.

Problem 2.5. (2021 Spring KAIST Calculus I) For the surface with equation

$$(y - \sqrt{2z_0})^2 - x^2 = z,$$

consider its curve cut C by the plane $z = z_0 > 0$.

(a) Identify the type of C and find its eccentricity.

(b) Find the foci and directrices of the curve cut on the plane $z = z_0$.

(c) Find the polar equation $r = f(\theta)$ of the curve cut on the plane $z = z_0$.

Problem 2.6. (2021 Fall KAIST Calculus I) Mark True or False for each of the following statements. You do not need to justify your answers.

(1) If a series $\displaystyle\sum_{n=0}^{\infty} a_n x^n$ converges on $[-1, 1]$, then the series $\displaystyle\sum_{n=1}^{\infty} n a_n x^{n-1}$ converges on $(-1, 1)$.

(2) If $\mathbf{a}, \mathbf{b}, \mathbf{c}$ are three-dimensional vectors, then

$$\mathbf{a} \times (\mathbf{b} \times \mathbf{c}) = (\mathbf{a} \times \mathbf{b}) \times \mathbf{c}.$$

(3) If $\mathbf{a}, \mathbf{b}, \mathbf{c}$ are nonzero three-dimensional vectors and $(\mathbf{a} \times \mathbf{b}) \cdot \mathbf{c} = 0$, then $\mathbf{a}, \mathbf{b}, \mathbf{c}$ are in the same plane.

(4) Assume f is infinitely differentiable at 0 (i.e., $f^{(n)}(0)$ exists for all $n = 1, 2, 3, \ldots$). Then there exists $\varepsilon > 0$ such that

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} x^n$$

for all $-\varepsilon < x < \varepsilon$.

(5) Let $A(x) = \displaystyle\sum_n a_n x^n$ and $B(x) = \displaystyle\sum_n b_n x^n$ be power series with radii of convergence R_1 and R_2, respectively. Then $A(x)B(x)$ is again a power series with radius of convergence $\min\{R_1, R_2\}$.

Problem 2.7. (2021 Fall KAIST Calculus I) Let $\{\mathbf{x}_n\}$ be a sequence of vectors, where

$$\mathbf{x}_n = (x_n^{(1)}, x_n^{(2)}, x_n^{(3)}).$$

Let $\mathbf{x}$ be a vector. Prove or disprove:

(a) $\|\mathbf{x}_n - \mathbf{x}\| \to 0$ if $\displaystyle\sum_{k=1}^{n} \|\mathbf{x}_k - \mathbf{x}\| \le 1$ for all n.

(b) $\|\mathbf{x}_n - \mathbf{x}\| \to 0$ if

$$\lim_{n\to\infty} \frac{\|\mathbf{x}_{n+1} - \mathbf{x}\|}{\|\mathbf{x}_n - \mathbf{x}\|} = \rho < 1.$$

Problem 2.8. (2021 Fall KAIST Calculus I) How many domains are separately enclosed by the following polar equations?

(1) $r = 4\cos(2\theta)$

(2) $r = \dfrac{4}{2 - \cos\theta}$

(3) $r = 2(1 + \cos\theta)$

Problem 2.9. (2021 Fall KAIST Calculus I) Consider the curve

$$r = \cos 3\theta, \qquad -\frac{\pi}{6} \le \theta \le \frac{\pi}{6}.$$

(a) Find the slopes of the curve at the origin.

(b) Find the length of the curve, using the fact that

$$\int_0^{\pi/3} \sqrt{1 + 8\sin^2 3\theta}\, \mathrm{d}\theta \approx 2.23.$$

(c) Find the area enclosed by the curve.

Problem 2.10. (2021 Fall KAIST Calculus I) Your eye is at $(4, 0, 0)$. You are looking at a triangular plate whose vertices are at $(1, 0, 1)$, $(1, 1, 0)$, and $(-2, 2, 2)$. The line segment from $(1, 0, 0)$ to $(0, 2, 2)$ passes through the plate. What portion of the line segment is hidden from your view by the plate?

Problem 2.11. (2021 Fall KAIST Calculus I) Let

$$L_1 : \ x = 1 + t, \ y = 3 - t, \ z = 2t, \qquad L_2 : \ x = 0, \ y = -2s, \ z = 5s.$$

Find the minimum of $\|\overrightarrow{P_1P_2}\|$ for $P_1 \in L_1$ and $P_2 \in L_2$.

3 (MATH311) Analysis I

4 (MATH312) Analysis II